

ATOM FACTORY — WORKED EXAMPLE

DISTRIBUTED MANUFACTURING ROADMAP

A four-phase production roadmap, derived from publicly available data, for the
SWCNT–TiO₂–Re Dye-Sensitized Oxygen Sensor

Based on: Wettstein et al., Advanced Science (2025)

DOI: 10.1002/advs.202405694 — ETH Zürich

Pilot: 5,000 sensors — Scale: 100,000 sensors

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Foreword: What This Document Demonstrates

Atom Factory is the manufacturing layer of ttOS. It uses the Contract Logic Engine (CLE) to engage university labs at scale through standardised consulting and services agreements, turning the global academic research infrastructure into a distributed production network for breakthrough materials.

This document is a worked example of the Atom Factory method, applied to a single published invention: the SWCNT–TiO₂–Re dye-sensitized oxygen sensor developed at ETH Zürich and published by Wettstein et al. in *Advanced Science* (2025). The complete roadmap that follows — from the original supplementary data, through a four-phase manufacturing plan, to a fully costed production model at 100,000 sensors per year — was constructed entirely from publicly available sources: the original paper, its supplementary information, the ETH Transfer technology sheet, and standard market data on materials, labour, and equipment.

It demonstrates what becomes possible when CLE-enabled contracts make it economical to commission production runs across twenty universities in parallel, when central CRO synthesis replaces twenty redundant in-lab batches, and when fabrication concentrates at low-cost nodes once the synthesis bottleneck is removed.

A note on status. This roadmap has not yet been executed. No sensors have been manufactured under the programme described here, no CRO contract has been placed, no university partner has been engaged. The document is a method demonstration, not a record of a delivered project. Atom Factory is currently seeking foundation funding and university partners to begin Phase 1 in 2026. The purpose of publishing this roadmap is to show, in full and defensible detail, what a real Atom Factory programme would look like — and to invite the conversations that would make it real.

The substance of the roadmap that follows is unchanged from the internal working version: every cost figure, every supplier name, every university capability assessment, and every risk has been carried over without simplification. What has been added is the framing above. What has been removed is nothing.

Executive Summary

This document tells the complete story of how a single university research paper — a supplementary data file from ETH Zürich describing a novel dye-sensitized oxygen chemiresistor — was systematically transformed into a three-phase global manufacturing roadmap.

The sensor in question is a remarkable device: a thin-film stack of single-walled carbon nanotubes (SWCNTs), amorphous titanium dioxide (TiO₂), and a rhenium carbonyl bipyridine sensitizer dye deposited on a gold interdigitated electrode. Under green visible light, it detects oxygen at concentrations as low as 157 parts per billion, is selective against all common interfering gases, tolerates humidity from 0 to 80%, and remains stable for over three months. It is, in short, a publishable result with real commercial potential.

The process that follows began with a careful read of the supplementary data: extracting which materials were commercially available and which had to be synthesised, understanding the synthesis bottleneck, mapping which universities worldwide could realistically execute the full process, and then building a rigorous cost model that distinguished university overhead, regional labour rates, material scale economies, and supply-chain risks.

The result is a three-phase roadmap:

Phase 1 ~\$120k <i>100 sensors, 20 universities</i>	Phase 2 ~\$42k <i>CRO synthesis, 100g Re dye</i>	Phase 3 (pilot) ~\$370k <i>5,000 sensors, India + Korea</i>
Phase 3 per sensor ~\$74 <i>vs ~\$1,200 at uni lab scale</i>	Electrode risk Critical <i>50% of Phase 3 cost</i>	Re supply risk Watch <i>\$6,112/kg and rising</i>

If Phase 3 pilot validation confirms use-case performance, the roadmap extends to full-scale production at 100,000 sensors per year, bringing the per-sensor cost to approximately \$14–\$20 depending on electrode sourcing.

1. Starting Point: The Supplementary Data

1.1 What the Paper Describes

The source paper, “A Dye-Sensitized Sensor for Oxygen Detection under Visible Light” (Wettstein et al., Adv. Sci. DOI 10.1002/advs.202405694), describes a three-component chemiresistor fabricated by depositing a composite film of SWCNTs and amorphous TiO₂ onto gold interdigitated electrodes, followed by sensitisation with a phosphonate-anchored rhenium carbonyl dye. The sensor responds to oxygen under illumination by green LED, with the excited-state rhenium complex acting as a photoredox mediator to shuttle charge between oxygen and the SWCNT network.

The supplementary information provided a complete synthetic and fabrication protocol, detailed characterisation data, and full sensing measurement protocols. This is the starting document for everything that follows.

1.2 Extracting the Input Materials

The first analytical task was to read the supplementary data and produce a clean inventory of every material required, categorised by whether it was commercially available off the shelf or required in-house synthesis or purification. The results are summarised in the table below.

Material	Commercially Available	Purification Required	Supplier
Ti(OiPr) ₄ 99.999%	Yes	No — use as received	Sigma-Aldrich 377996
[Re(CO) ₅ Br]	Yes	No — use as received	Sigma-Aldrich 327638
P3-SWNT (>90% purity)	Yes	No — dry under vacuum only	Carbon Solutions Inc.
4,4'-Dibromo-2,2'-bipyridine	Yes	No	Chem-Impex / abcr
PPh ₃ 99%	Yes	No	Sigma-Aldrich
Pd(PPh ₃) ₄ 99%	Yes	No	Sigma-Aldrich 216666
Diethyl phosphite	Yes	No	Sigma-Aldrich
Bromotrimethylsilane 97%	Yes	No (moisture-sensitive)	Sigma-Aldrich 194409
Et ₃ N	Yes	No	Sigma-Aldrich
Anhydrous DMSO	Yes	No — use Sure/Seal	Sigma-Aldrich 276855

Material	Commercially Available	Purification Required	Supplier
Anhydrous toluene	Yes	No — use Sure/Seal	Sigma-Aldrich 244511
Anhydrous CHCl ₃ , MeOH, EtOH	Yes	No — use Sure/Seal	Sigma-Aldrich
IDEAU200 gold IDE electrodes	Yes	No	Metrohm DropSens DRP-IDEAU200-U50
N3 dye (control only)	Yes	Yes — recrystallisation	Solaronix / Ossila
N719 dye (control only)	Yes	Yes — recrystallisation	Solaronix / Ossila
[(Pbpy)(CO) ₃ ReBr] Re sensitizer	NO	N/A — must be synthesised	No commercial source exists

1.3 The Synthesis Bottleneck

Critical finding: [(Pbpy)(CO)₃ReBr] has no commercial source anywhere in the world. Every sensor requires this compound to be synthesised from scratch. This is the single most important constraint in the entire manufacturing chain.

Every other input can be purchased from a catalogue and used as received. The rhenium sensitizer dye — the compound that gives the sensor its visible-light response — does not exist commercially. It must be made in a two-step synthesis starting from [Re(CO)₅Br] and 4,4'-dibromo-2,2'-bipyridine.

This synthesis has three significant requirements that immediately constrain which facilities can perform it:

- An argon-filled glovebox (or at minimum a well-maintained Schlenk line) is required for the [Re(CO)₅Br] complexation step, which is conducted under strictly inert atmosphere conditions.
- A Pd-catalysed phosphonate coupling (Pd(PPh₃)₄, 110°C reflux in toluene, 15 hours under argon) is required for the first step, demanding competent Schlenk or glovebox technique.
- Purification requires flash chromatography (silica, 0–4% MeOH in DCM gradient), followed by a silyl deprotection step, a precipitation sequence, and high-vacuum drying.

The overall yield of the two-step sequence is approximately 29% (44% for Step 1, 65% for Step 2). This yield factor directly sets the amount of starting material that must be procured for any given production target.

The Two-Step Synthesis in Brief

Step	Reaction	Conditions	Yield
Step 1	Pd-catalysed phosphonate coupling of dibromo-bipyridine	Pd(PPh ₃) ₄ , Et ₃ N, toluene, 110°C, 15h, argon	~44%
Step 2a	Complexation with [Re(CO) ₅ Br]	1:1, dry toluene, 80°C, 19h (microwave or oil bath)	~94%
Step 2b	Silyl deprotection with bromotrimethylsilane	TMSBr (68 eq), CHCl ₃ , 65°C, 24h, then MeOH	~65%
Overall	4,4'-dibromo-bpy → [(Pbpy)(CO) ₃ ReBr]	Two steps, one purification	~29%

Confirmation of successful synthesis requires ¹H NMR (DMSO-d₆: δ 9.17, 8.75, 7.92 ppm), ³¹P NMR (δ 5.56 ppm), FT-IR (CO bands at 2033 and 1915 cm⁻¹), and UV-vis (DMF: λ_{max} 301 nm, λ_{MLCT} 389 nm). This characterisation is non-negotiable: without it, the active dye cannot be confirmed and sensor performance will be unpredictable.

2. Who Can Make This? The University Capability Map

2.1 The Two-Tier Problem

The synthesis bottleneck immediately creates a two-tier manufacturing challenge. Batch 1 — the Re sensitizer synthesis — requires a fully equipped inorganic/organometallic chemistry laboratory: glovebox or Schlenk line, microwave or high-temperature reflux capability, flash chromatography, high vacuum, and NMR access. Batch 2 — electrode fabrication — requires no glovebox at all; it needs only an airbrush, a heating plate, a syringe pump, an ultrasonic bath, and basic electrical testing equipment.

This split means that Tier 1 institutions (full synthesis capability) are required for Phase 1, while Tier 2 and Tier 3 institutions can handle Phase 3 fabrication once centralised Re sensitizer supply is established.

Tier	Required Capability	Suitable For
Tier 1	Glovebox or Schlenk, Pd catalysis, flash chromatography, NMR, FT-IR	Phase 1 (Batch 1 + Batch 2)
Tier 2	Airbrush, heating plate, syringe pump, ultrasonic bath, resistance meter	Phase 3 (Batch 2 only)
Tier 3	Basic wet lab and electrical testing	Phase 3 QC and assembly only

2.2 Verification Strategy

There is no global database that catalogues every piece of equipment at every university chemistry department. However, a practical three-tier verification strategy was developed that can confirm the full list of ~80 qualifying institutions within two to three working days.

For approximately 60 institutions — all G20-country research universities with confirmed organometallic publication output in the last three years — the assumption of full Tier 1 capability is well-founded and requires no further verification. If a group has published a paper in the last three years involving air-sensitive organometallic synthesis, they have the equipment. For a further 15 institutions where the primary research focus is materials science rather than organometallic chemistry, a ten-minute Google Scholar search confirms whether a relevant group exists. For the remaining 5 institutions (primarily newer South American or Central/Eastern European universities), a single email to the departmental facilities coordinator resolves the question definitively.

2.3 The Global Tier 1 University List

The following institutions were identified as having confirmed or near-certain Tier 1 capability for Phase 1 participation. Approximately 80 institutions qualify globally; 20 will be selected for Phase 1.

Region	Institutions (selected examples)	Count
USA	MIT, Caltech, UC Berkeley, Stanford, Harvard, UIUC, Northwestern, UCLA, Georgia Tech, Columbia, UChicago, Yale, Princeton, U Michigan, Johns Hopkins	21
Canada	U Toronto, McGill, UBC, U Alberta, McMaster, U Waterloo	6
UK	Oxford, Cambridge, Imperial, UCL, Manchester, Edinburgh, Bristol, York, Warwick, Birmingham, Nottingham, Liverpool, St Andrews, Glasgow	14
Switzerland	ETH Zürich, EPFL, U Basel, U Zurich	4
Germany	LMU Munich, TU Munich, Heidelberg, RWTH Aachen, Würzburg, Münster, TU Dresden, Göttingen, FAU Erlangen, Freiburg, Mainz, Ruhr Bochum	14
France / Benelux / Scandinavia	Paris-Saclay, Sorbonne, Strasbourg; KU Leuven, Ghent; TU Delft, Groningen, Utrecht; Uppsala, Stockholm, Lund, Copenhagen, DTU, Aarhus	16
India	IIT Bombay, IIT Madras, IIT Delhi, IIT Kanpur, IIT Kharagpur, IISc Bangalore, IISER Pune	7
South Korea	KAIST, Seoul National University, POSTECH, Yonsei, SKKU	5
Singapore / Australia	NUS, NTU; ANU, U Melbourne, U Sydney, U Queensland	6
South America / CEE	USP, UNICAMP, UBA Argentina; IOCB Prague, Polish Academy of Sciences	5

3. The Cost Model: Labour Rates and Charge Rates

3.1 Regional Labour Rates

University production costs are dominated by labour. The effective hourly rate for a postdoctoral researcher — the appropriate skill level for this synthesis and fabrication work — varies by approximately 15-fold between the most expensive and least expensive regions in the qualified university list. The following rates include a 20–30% facilities and administrative (F&A) overhead applied to labour, consistent with typical university research project accounting.

Region	Postdoc Annual Salary	Effective Hourly Rate (incl. OH)	Cost Tier
United States	~\$61,000/yr (NIH NRSA minimum)	~\$37/hr	High
Switzerland (ETH/EPFL)	~CHF 92,500/yr (~\$99,800 USD)	~\$60/hr	Highest
Germany	~€59,000/yr (~\$64,000 USD)	~\$40/hr	High
United Kingdom	~£34,000/yr (~\$43,000 USD)	~\$26/hr	Mid
South Korea	~₩50,600,000/yr (~\$37,000 USD)	~\$23/hr	Mid-Low
India (IIT/IISc)	~₹660,000/yr (~\$7,900 USD)	~\$5.70/hr	Low

3.2 Equipment and Facility Charge Rates

In addition to labour, core facility instruments carry their own charge rates. The most significant for this production process are NMR spectrometers, which are required to confirm successful Re sensitizer synthesis. The following rates represent typical internal academic user pricing.

Instrument	Internal Academic Rate	Relevance to This Process
NMR 400–600 MHz (solution)	\$10–\$25/hr	Required: ^1H + ^{31}P confirmation of Re dye
FT-IR spectrometer	~\$5–15/run (consumables only)	Required: CO bands 2033 and 1915 cm^{-1}
UV-vis spectrophotometer	~\$5–10/run	Required: λ_{MLCT} 389 nm confirmation

Instrument	Internal Academic Rate	Relevance to This Process
Flash chromatography (Biotage)	Included in OH or ~\$20–50/run	Required: Step 1 purification
High vacuum pump	Included in lab overhead	Required: product drying between steps
Airbrush + heating plate	~\$80–\$400 capital (amortised)	Batch 2 spray-coating of electrodes
Resistance meter / DAQ	~\$0.50/electrode	QC check: target 1–5 k Ω per device

4. Phase 1 — Distributed University Pilot: 100 Sensors across 20 Universities

4.1 Design of the Pilot

Phase 1 serves three purposes: it validates that the ETH Zürich protocol transfers faithfully to other institutions; it generates 100 sensors for use-case testing and application validation; and it begins building the international network of fabrication expertise that will underpin Phase 3.

The design calls for 20 universities — selected from the 80 in the qualified list — to each perform the complete synthesis and fabrication process independently. Each site produces 5 finished sensors (plus one spare to absorb yield loss) from a single Batch 1 synthesis run and a single Batch 2 fabrication run.

The key inefficiency of Phase 1 is deliberate: each site running its own Batch 1 synthesis generates redundant overhead but maximises protocol learning and independence. This is correct for a validation phase.

4.2 Sensor Production per Site

- One complete Batch 1 Re sensitizer synthesis (6 person-days: two multi-day reflux steps, one column chromatography, one silyl deprotection, high-vacuum drying, NMR/IR/UV-vis confirmation)
- One Batch 2 fabrication run producing 6 SWCNT-TiO₂-Re electrodes (2 person-days: SWCNT-TiO₂ preparation, 4-week aging in water, spray-coating, overnight drying, sensitizer soaking, vacuum drying)
- QC: resistance testing all 6 electrodes, targeting 1–5 kΩ; pass threshold ~80%, giving 5 shippable sensors per site

4.3 Cost by Region

Region Group	Universities	Cost per Site (5 sensors)	Subtotal
US / Switzerland (Tier 1 highest)	8	\$4,200–\$4,800	~\$35,200
Germany / UK / Netherlands (Tier 1 mid)	6	\$2,800–\$3,200	~\$18,000
South Korea — KAIST, SNU, POSTECH (Tier 1)	3	\$2,200–\$2,600	~\$7,200

Region Group	Universities	Cost per Site (5 sensors)	Subtotal
mid-low)			
India — IIT Bombay, IIT Madras, IISc (Tier 1 low)	3	\$1,400–\$1,800	~\$4,800
TOTAL — 20 universities	20	blended ~\$3,400	~\$68,000

4.4 Phase 1 Per-University Cost Breakdown (Mid Estimate)

Cost Item	Amount	Notes
Batch 1 materials (Re synthesis reagents)	~\$708	Fixed regardless of output quantity
Batch 2 materials (6 IDEAU200 electrodes + chemicals)	~\$132	\$10/electrode × 6 + solvents, consumables
Synthesis labour (6 person-days)	\$380–\$1,840	Varies by regional rate
Fabrication labour (0.4 person-days)	\$25–\$120	Varies by regional rate
NMR + FT-IR + UV-vis confirmation	~\$80	Internal academic facility rates
University F&A overhead (30% on labour)	\$120–\$590	Applied to all labour
Consumables and contingency	~\$50	Glassware, filters, Ar gas usage

Per-sensor cost at Phase 1: ~\$680. This is high because a full Batch 1 synthesis (6 person-days, \$708 in reagents) is amortised over only 5 sensors per site. Eliminating this redundancy is the entire rationale for Phase 2.

4.5 Phase 1 Deliverables and Exit Criteria

- 100 confirmed sensors (resistance 1–5 kΩ, UV-vis λMLCT 389 nm confirmed) distributed to use-case testing partners
- Protocol transfer confirmed at ≥15 of 20 sites (i.e. ≥75% yield on Batch 1 confirmation by NMR/IR)
- First use-case data: response to target oxygen concentrations under green LED, selectivity vs. interfering gases, humidity tolerance
- Identification of 3–5 low-cost fabrication nodes (India, Korea) for Phase 3 scale-up

- Recommendation on OEM electrode substitution based on parallel validation experiment (200 test electrodes from Chinese OEM supplier)

5. Phase 2 — CRO Bulk Synthesis of the Re Sensitizer

5.1 The Strategic Logic

Phase 1 reveals the fundamental economics problem of distributed synthesis: each of 20 universities runs an independent full Batch 1 synthesis, each burning 6 person-days of skilled organometallic chemist time and \$708 in reagents to produce enough Re sensitizer for 5 sensors. The total Phase 1 labour cost for synthesis alone, across all 20 sites, is approximately \$30,000–\$40,000. This is inherently inefficient.

Phase 2 solves this by outsourcing all Batch 1 synthesis to a single specialist contract research organisation (CRO). One well-resourced synthetic chemistry CRO, running the two-step process at 100 gram scale, produces enough $[(Pbpy)(CO)_3ReBr]$ for approximately 2 million sensor soaking doses — more than sufficient for Phase 3 at 5,000 sensors, Phase 4 at 100,000 sensors, and significant reserve. The 20 parallel university synthesis runs, each with their own overhead and variability, are replaced by a single quality-controlled batch with full analytical certification.

5.2 CRO Cost Model — 100g Final Product

Item	Quantity / Basis	Cost Estimate
$[Re(CO)_5Br]$ precursor (140g required at 29% yield)	Bulk procurement, ~\$90/g at 100g+ volume	~\$12,600
4,4'-Dibromo-2,2'-bipyridine (~100g batch)	Bulk pricing ~\$80/g	~\$8,000
$Pd(PPh_3)_4$ catalyst (gram-scale amount)	Bulk price	~\$2,500
PPh_3 , diethyl phosphite, bromotrimethylsilane, Et_3N	Bulk solvents and reagents	~\$600
Anhydrous solvents: toluene, $CHCl_3$, MeOH (bulk drums)	Industrial grade, pre-dried	~\$500
CRO labour (~3 chemist-weeks, organometallic specialty)	FTE rate at European CRO	~\$9,000–\$14,000
Purification (flash chromatography, precipitation, QC)	NMR, IR, UV-vis, HPLC certification	~\$2,500
Argon-sealed packaging + cold-chain shipping to nodes	Multi-site distribution	~\$500
TOTAL CRO INVOICE		~\$36,000–\$46,000

At approximately \$420/g, the CRO route is 65% cheaper than university-scale production at ~\$1,200/g. More importantly, it delivers a homogeneous, fully characterised batch with CoA (certificate of analysis), eliminating the inter-site variability that would otherwise affect Phase 3 yield and performance.

5.3 Recommended CRO Partners

CRO	Location	Relevant Capability	Recommendation
Taros Discovery	Germany (+ Asia pilot plant)	Organometallic, inert atmosphere, pilot scale	Priority recommendation
Rieke Metals	USA	40+ years air-sensitive organometallic, inert atmosphere standard	Priority recommendation
ChiroBlock	Germany	Fixed-fee-for-service guaranteed delivery model	Good for fixed budget
Enamine	EU / Ukraine	Large FTE capacity, lower cost, broad chemistry	Verify Re capability first
Life Chemicals	EU	Materials science track record	Confirm organometallic depth

5.4 Critical Supply Risk: Rhenium Metal

WARNING: Rhenium metal price has risen +206% since January 2024 and +146% in 2025 alone, reaching \$6,112/kg in March 2026. It is now classified as a US critical mineral. A fixed-price CRO contract should be placed immediately.

Despite this dramatic price spike, the rhenium metal content per finished sensor is only approximately 0.4mg — a metal cost of approximately \$0.002 per sensor, which is negligible. The real financial risk is the $[\text{Re}(\text{CO})_5\text{Br}]$ precursor cost for the Phase 2 CRO run. At current prices, the 140g needed costs approximately \$12,600–\$21,000. If rhenium prices double again to \$12,000/kg, the precursor alone would cost \$36,000.

Mitigation Action	Priority	Saving Potential
Fix-price CRO contract locking in $[\text{Re}(\text{CO})_5\text{Br}]$ cost now	Priority 1 — act immediately	Avoids unknown upside cost
Bulk-buy $[\text{Re}(\text{CO})_5\text{Br}]$ (500g–1kg)	Priority 2	\$20k–\$40k saved if prices double

Mitigation Action	Priority	Saving Potential
upfront as 12-month buffer		
CRO to synthesise $[\text{Re}(\text{CO})_5\text{Br}]$ from $\text{Re}_2(\text{CO})_{10} + \text{Br}_2$ directly	Priority 3	~30% reduction in precursor cost
Monitor US USGS critical mineral re-classification	Ongoing watch	Export control risk if Re designated strategic

6. Phase 3 — Pilot Scale: 5,000 Sensors at Low-Cost Nodes

6.1 The Phase 3 Premise

Phase 3 is a pilot production run of 5,000 sensors, triggered by successful validation of Phase 1 use cases. It rests on three structural changes from Phase 1:

First, all Re sensitizer is supplied centrally from the Phase 2 CRO run. No site in Phase 3 performs any Batch 1 synthesis. The glovebox requirement disappears entirely from Phase 3. Every site in Phase 3 only needs to do Batch 2: SWCNT-TiO₂ preparation, spray-coating, and sensitizer soaking.

Second, fabrication is concentrated exclusively at low-cost nodes. India (IIT Bombay, IIT Madras, IISc Bangalore) and South Korea (KAIST, SNU, POSTECH) are the primary Phase 3 sites, with Indian labs taking 60% of volume and Korean labs taking 40%. The 15-fold difference in effective hourly rate between Switzerland and India means that concentrating Batch 2 at Indian and Korean nodes produces dramatic per-sensor cost reduction.

Third, materials are procured in bulk. Electrode unit prices at 10,000+ units are approximately 50–65% below the retail price of a pack of 50. SWCNT bulk pricing at 20g scale is approximately 70% below per-gram research pricing.

6.2 Phase 3 Full Cost Model — 5,000 Sensors

Total Phase 3 cost ~\$370,000 <i>materials + labour + logistics</i>	Per sensor cost ~\$74 <i>97% reduction vs Phase 1</i>	Material cost ~\$265,000 <i>electrodes dominate at 56%</i>
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Cost Item	Basis	Cost
IDEAU200 gold IDE electrodes ×6,250 (25% yield buffer)	Bulk OEM pricing ~\$3.50–\$5.00/unit	~\$22,000–\$31,000
[(Pbpy)(CO) ₃ ReBr] — amortised from Phase 2 CRO run	100g covers all of Phase 3 with large reserve	~\$42,000 (full Phase 2 cost)
P3-SWNT (~1g total at 0.2mg/sensor)	Bulk pricing ~\$150/g at this quantity	~\$150
Ti(OiPr) ₄ — small quantity, bulk drum shared	Fractional use of 5L drum	~\$200
Anhydrous DMSO, EtOH,	Bulk drums at industrial pricing	~\$3,000

Cost Item	Basis	Cost
solvents — bulk industrial grade		
Nylon membranes, crimp vials, airbrush consumables, Ar gas	100 fabrication batches × 50 sensors	~\$4,000
QC: resistance testing, spot-check UV-vis, 5% sampling	In-lab, low marginal cost	~\$1,500
Cold-chain logistics: Re complex to nodes, sensors outbound	International freight, temperature controlled	~\$4,500
Labour — India node (60%): 120 person-days at ~\$45/day	IIT/IISc postdoc effective rate	~\$5,400
Labour — Korea node (40%): 80 person-days at ~\$72/day	KAIST/SNU postdoc effective rate	~\$5,760
QC coordination overhead (20% on all labour)	Supervision, protocol adherence	~\$2,230
TOTAL		~\$90,000–\$99,000*

* Note: The Re sensitizer line item of ~\$42,000 represents the full cost of the Phase 2 CRO run. Amortised across Phase 3 only, it amounts to ~\$8.40/sensor. If Phase 2 is treated as a sunk cost prior to Phase 3 launch, the incremental Phase 3 cost is approximately \$48,000–\$57,000, or ~\$9.60–\$11.40/sensor. The full economic model shown above (\$90k–\$99k) includes the Phase 2 Re synthesis cost in full to give a complete view of total programme investment.

6.3 The Electrode Cost Lever

The electrode substrate is 50%+ of Phase 3 material cost. Replacing branded DropSens IDEAU200 with a validated custom OEM equivalent is the single highest-leverage cost reduction available in the entire programme.

Electrode Option	Unit Price	Total for 6,250 units	Validation Required
DropSens IDEAU200 (branded, retail)	~\$10/unit	~\$62,500	None — reference standard
DropSens IDEAU200 (bulk OEM, 10k+ order)	~\$3.50–\$5.00/unit	~\$22,000–\$31,000	Confirm pricing with Metrohm
Chinese OEM custom IDE (Al ₂ O ₃ ceramic, gold, 100µm pitch)	~\$0.80–\$1.50/unit	~\$5,000–\$9,375	200-unit geometry validation experiment required

A 200-unit geometry validation experiment costs approximately \$3,000 and should be run in parallel during Phase 1. If the custom OEM electrode is confirmed equivalent, the Phase 3 electrode cost is reduced by \$17,000–\$26,000, and the per-sensor cost falls to approximately \$65–70. This experiment is strongly recommended before placing Phase 3 electrode procurement.

6.4 Phase 3 Labour Model

Node	Volume	Fabrication Batches	Person-Days	Labour Cost
India (IIT Bombay, IIT Madras, IISc)	3,000 sensors (60%)	60 batches × 50 sensors	120 days	~\$5,400
South Korea (KAIST, SNU, POSTECH)	2,000 sensors (40%)	40 batches × 50 sensors	80 days	~\$5,760
QC coordination overhead (20%)	—	—	40 days effective	~\$2,230
TOTAL	5,000 sensors	100 batches	240 person-days	~\$13,390

7. Use Case Validation and the Phase 4 Decision Gate

7.1 What Validation Must Confirm

The decision to proceed from Phase 3 (5,000 sensors) to Phase 4 (100,000 sensors) is gated on use-case validation, not just protocol transfer. The sensor is rated at Technology Readiness Level 5 (TRL 5) by ETH Transfer (Reference 2024-034): a prototype has been built and tested in the lab, with a patent pending. What must now be confirmed is whether it performs well enough in real deployment environments to justify a \$1.4M production commitment. The ETH transfer sheet and the published paper together define seven specific target application domains, each with distinct validation requirements.

The ETH transfer sheet (Reference 2024-034) and the published paper together identify seven specific target application domains. These are the segments against which Phase 3 sensors must be validated:

Application Domain	Target O ₂ Range	Key Challenge to Validate	Success Threshold
Exhaust gas analysis — automobile and industrial process control	ppm–% O ₂ (exhaust: 0.1–15%)	Confirmed selectivity vs. CO, CO ₂ , CH ₄ , C ₂ H ₄ (paper Fig. 5A). Validate at elevated temperature and soot/particulate exposure in real exhaust stream.	No cross-response to CO/CO ₂ /hydrocarbons; stable at 50–200°C
Water and air quality monitoring	0.1 ppm – 21% O ₂	Dissolved O ₂ equivalent requires aqueous-phase adaptation. Air quality: confirmed humidity tolerance 0–80% RH (paper Fig. S32). Low-power field deployment <0.6 W (ETH transfer sheet).	Stable >6 months outdoor; operation at 0–80% RH confirmed
Early detection of food spoilage (modified atmosphere packaging)	100 ppm – 2% O ₂ (headspace)	Must operate through or integrated into packaging; temperature range 2–25°C. Sensitivity at low ppm levels confirmed (LOD 2.2 ppm at 1 min exposure, paper Fig. 4).	LOD ≤5 ppm; stable at 2–25°C; no response to CO ₂ , C ₂ H ₄ (spoilage gases)
Breath monitoring and personal health	15–21% O ₂ (exhaled breath: ~16%)	Sensor operates under ambient O ₂ with confirmed sensitivity to concentration changes of ≥26 ppm (paper Fig. 6). Miniature design and <0.6 W power confirmed. Biocompatibility not yet assessed.	Detects ≥26 ppm ΔO ₂ change under ambient air; miniature form factor; <0.6 W
Personnel health	16–21% O ₂	Sensitivity to O ₂ concentration	Response <10 s to

Application Domain	Target O ₂ Range	Key Challenge to Validate	Success Threshold
protection (confined spaces, industrial O ₂ depletion alarm)	(depletion alert at <18%)	decreases confirmed under ambient air carrier gas (paper Fig. 6). Rapid t ₉₀ = 6 s response is suitable for alarm applications. Must validate battery-powered operation.	1% O ₂ drop; stable >3 months; operable from small battery
Food and pharmaceutical process control (inert gas purity, N ₂ blanket)	20–500 ppm O ₂ (trace monitoring)	Confirmed selectivity over N ₂ O, CO ₂ , H ₂ , CH ₄ , N ₂ (paper Fig. 5A). TRL 5 as per ETH transfer sheet — prototype built and lab-tested. Must validate against regulatory standards (e.g. pharmacopoeia O ₂ limits).	LOD <200 ppb (paper S30); no cross-response; stable >3 months ambient
Cell health and disease progression monitoring (biomedical research)	1–200 ppm O ₂ (hypoxia: <20 ppm dissolved equivalent)	Paper cites cell health monitoring as target application. Aqueous-phase adaptation required. ppb-level LOD (157 ppb at 60 min) relevant for hypoxia detection. Biocompatibility study required.	LOD 157 ppb at 60 min exposure; selectivity vs. biological interferants to be assessed

7.2 The 4-Week Aging Observation

A key finding from the paper: SWCNT-TiO₂-Re electrodes improve significantly after 4 weeks of aging in water dispersion prior to sensitization. Ostwald ripening increases TiO₂ surface area and dye loading. This 4-week wait is non-negotiable and must be built into all Phase 3 and Phase 4 production schedules.

7.3 Phase 4 Decision Gate Criteria

Phase 4 (100,000 sensors) should be authorised when all of the following are confirmed from Phase 3 data:

- At least one use case with a confirmed commercial partner and purchase intent for a minimum of 50,000 sensors/year
- Field performance matching laboratory specifications to within 20% across at least two independent deployment sites
- Six-month field stability data (at least 80% of Phase 3 sensors still within spec at month 6)
- Custom OEM electrode validated as equivalent to DropSens IDEAU200

- Phase 2 CRO has demonstrated capacity and willingness to supply at Phase 4 volume (multiple 100g runs per year)
- Rhenium supply secured via fixed-price contract covering at least 18 months of Phase 4 production

8. Phase 4 — Full Scale: 100,000 Sensors per Year

8.1 Phase 4 Cost Model

Phase 4 assumes all Phase 3 validations have passed: OEM electrode is confirmed, Re supply is secured at fixed price, and at least one commercial partner with high-volume purchase intent is in place. Fabrication remains at India and Korea nodes, with the possibility of adding one additional low-cost node (e.g. UNICAMP Brazil or SKKU Korea) to handle volume and provide redundancy.

Phase 4 total cost ~\$1.38M <i>100,000 sensors/year</i>	Per sensor (OEM electrode) ~\$9.50 <i>if custom IDE validated</i>	Per sensor (DropSens bulk) ~\$13.80 <i>if branded electrode retained</i>
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Cost Item	Basis	Phase 4 Cost
Gold IDE electrodes ×125,000 (20% yield buffer)	Custom OEM ~\$1.20/unit OR bulk DropSens ~\$4.00/unit	\$150,000 OR \$500,000
[(Pbpy)(CO) ₃ ReBr] — multiple CRO batches	~3 × 100g runs/year at \$42k each	~\$126,000/year
P3-SWNT ~20g total (bulk)	~\$150/g at 20g scale	~\$3,000
Ti(OiPr) ₄ , solvents, bulk industrial grade	5L drum + bulk DMSO/EtOH	~\$13,600
Consumables (membranes, vials, Ar, airbrush)	2,000 batches × 50 sensors	~\$48,000
QC sampling protocol (5%)	Resistance + spot-check UV-vis per 20 batches	~\$15,000
Cold-chain logistics (Re in, sensors out)	Annual programme	~\$18,000
Labour — India (60%): 2,400 person-days at \$45/day	IIT/IISc postdoc effective rate	~\$108,000
Labour — Korea (40%): 1,600 person-days at \$72/day	KAIST/SNU postdoc effective rate	~\$115,200
QC coordination (20% on labour)		~\$44,640
TOTAL (OEM electrode) / (DropSens bulk)		\$641,240 / \$991,240

8.2 Phase 4 Scale Economics Summary

Scenario	Volume	Per-Sensor Cost	Key Assumption
Phase 1 (university distributed)	100 sensors	~\$680/sensor	20× parallel Batch 1 synthesis
Phase 3 pilot (CRO Re supply)	5,000 sensors	~\$74/sensor (incl. Phase 2)	Low-cost nodes only, no glovebox in Batch 2
Phase 3 incremental only	5,000 sensors	~\$10–\$12/sensor	Phase 2 Re synthesis treated as sunk cost
Phase 4 — OEM electrode validated	100,000 sensors	~\$9.50/sensor	Custom IDE + CRO Re at \$42k/batch
Phase 4 — DropSens bulk pricing	100,000 sensors	~\$13.80/sensor	Branded electrode retained at bulk OEM price

9. Risk Register

Risk	Probability	Impact	Mitigation
Re metal price continues rising beyond \$10,000/kg	Medium	High — adds \$20k–\$40k to CRO run	Fix-price CRO contract; bulk buy [Re(CO) ₅ Br] now
US/USGS export controls on Re as strategic mineral	Low-Medium	Very High — supply chain disruption	Source [Re(CO) ₅ Br] from non-US CRO (Germany, India)
Protocol fails to transfer at some Phase 1 sites	Medium	Low — affects <20 sensors	Peer-to-peer protocol sharing; ETH Zürich advisory
OEM electrode fails geometry validation	Low-Medium	Medium — raises Phase 3/4 cost	Validate in Phase 1 before placing Phase 3 order
4-week aging step forgotten at Phase 3 nodes	Medium	High — batch of sensors underperforms	Written SOP, mandatory; QC check UV-vis λ MLCT before soaking
SWCNT batch variability (P3-SWNT lot-to-lot)	Low	Medium — resistance variation in spray-coating	Buy 1-year supply in single lot; lot qualification test
Use case fails field validation — Phase 4 not triggered	Medium	High — strands 5,000 sensors and Phase 2 investment	Phase 3 sensors are functional devices with real LOD 157 ppb; multiple use cases reduce single-point dependency
Competing commercial sensor enters market pre-Phase 4	Medium	Medium — price pressure	Visible-light operation and ppb-range LOD are differentiating; patent analysis recommended

10. Integrated Programme Roadmap

Phase	Activity	Timeline	Cost	Gate to Next Phase
Phase 1	100 sensors across 20 universities; protocol transfer; use-case testing; OEM electrode validation	Months 1–12	~\$68,000	Protocol transfer at ≥15/20 sites; use-case data from ≥2 applications
Phase 2	CRO contract for 100g [(Pbpy)(CO) ₃ ReBr]; fix-price Re metal; CoA delivery	Months 4–8 (parallel)	~\$42,000	Full CoA: NMR, IR, UV-vis, HPLC; argon-sealed delivery to nodes
Phase 3	5,000 sensors at India + Korea nodes; Batch 2 only; centralised Re supply	Months 9–18	~\$90,000–\$99,000	6-month field stability; commercial partner purchase intent; OEM electrode decision
Phase 4	100,000 sensors/year; full programme; OEM electrode if validated	Months 18+	~\$641k–\$991k/year	Ongoing commercial programme

Total programme investment to Phase 4 entry: ~\$200,000–\$210,000 (Phases 1+2+3 combined). This is the validation budget required before committing to \$641k–\$991k/year in volume production.

Appendix A: Complete Material and Equipment Lists

A.1 Batch 1 — Re Sensitizer Synthesis (per run)

Material	Sigma-Aldrich / Supplier Cat. No.	Unit Price	Amount per Run
Ti(OiPr) ₄ 99.999%	Sigma 377996	~\$80/25mL	50 µL
[Re(CO) ₅ Br]	Sigma 327638	~\$150/1g	1.0 eq (~1g)
4,4'-Dibromo-2,2'-bipyridine	Chem-Impex / abcr	~\$133/1g	1.0 eq
Pd(PPh ₃) ₄ 99%	Sigma 216666	~\$95/1g	0.05 eq
PPh ₃ 99%	Sigma T84409	~\$35/100g	5.0 eq
Diethyl phosphite	Sigma	~\$30/100mL	2.3 eq
Bromotrimethylsilane 97%	Sigma 194409	~\$45/25mL	68 eq
Et ₃ N	Sigma	~\$30/100mL	2.3 eq
Anhydrous toluene Sure/Seal	Sigma 244511	~\$40/100mL	60 mL
Anhydrous CHCl ₃ Sure/Seal	Sigma 288306	~\$40/100mL	As needed
Anhydrous MeOH Sure/Seal	Sigma 322415	~\$35/100mL	3 mL + washes
Silica gel for chromatography	Sigma	~\$30/500g	Column prep

A.2 Batch 2 — Electrode Fabrication (per 50-electrode run)

Material	Supplier	Unit Price	Amount per Run
IDEAU200 gold IDE electrodes	Metrohm DropSens DRP-IDEAU200-U50	~\$10/unit	62 units (25% yield buffer)
P3-SWNT >90%	Carbon Solutions Inc.	~\$280/g (research); ~\$150/g (bulk)	~0.01g (10 mg)
Ti(OiPr) ₄ 99.999%	Sigma 377996	~\$80/25mL	50 µL
Anhydrous DMSO Sure/Seal	Sigma 276855	~\$45/100mL	2 mL per electrode soak

Material	Supplier	Unit Price	Amount per Run
Anhydrous EtOH Sure/Seal	Sigma 459836	~\$35/100mL	Dispersion and washes
[(Pbpy)(CO) ₃ ReBr] (Phase 2 supply)	CRO Phase 2	~\$420/g (CRO)	~0.5 mL of 0.2 mM DMSO
Nylon membrane filters 0.2 µm	Sigma / Merck	~\$0.50/filter	1 per run
Crimp vials, Ar gas, consumables	Lab supply	~\$30–40/run	—

A.3 Equipment List — Tier Requirements

Equipment	Required Tier	Estimated Cost	Notes
Argon-filled glovebox	Tier 1 only	\$30,000–\$80,000	Required for Batch 1 complexation step
Schlenk line / vacuum manifold	Tier 1 only	\$2,000–\$5,000	Alternative to glovebox for experienced operators
Microwave reactor (Biotage Initiator+)	Tier 1 (or oil bath)	\$20,000–\$40,000	Conventional oil bath acceptable for Step 2a
Rotary evaporator	Tier 1 only	\$3,000–\$8,000	Required for solvent removal between steps
Flash chromatography (Biotage Isolera)	Tier 1 only	\$15,000–\$30,000	Required for Step 1 purification
High vacuum pump	Tier 1 only	\$1,500–\$5,000	Required for product drying (overnight under HV)
NMR spectrometer 300–500 MHz	Tier 1 (core facility)	\$300k–\$800k or facility	¹ H + ³¹ P required for batch confirmation
FT-IR spectrometer	Tier 1 (in-lab)	\$15,000–\$40,000	CO bands at 2033 and 1915 cm ⁻¹ required
Airbrush (Iwata Revolution BR)	Tier 2	~\$80	Batch 2 spray-coating — no special facility needed
Heating plate (150°C capable)	Tier 2	\$300–\$800	Batch 2 spray substrate temperature
Syringe pump	Tier 2	\$500–\$1,500	Controlled water addition to SWCNT-TiO ₂ mixture
Ultrasonic bath	Tier 2	\$500–\$3,000	SWCNT-TiO ₂ dispersion preparation
Agilent DAQ970A +	Tier 2	\$3,000–\$6,000	Resistance measurement of

Equipment	Required Tier	Estimated Cost	Notes
multiplexer			finished electrodes
Green LED (LCFOCUS LC-10RGB-C30)	Tier 2	~\$50	Sensing activation (200 mW/mm ² at 2cm)

Appendix B: Sensor Performance Specifications

Parameter	Value	Notes
Limit of detection (LOD)	157 ppb O ₂	Under green LED illumination, ambient conditions
Detection range	20 ppm – 200,000 ppm (20%)	Linear range in chemiresistive response
Response time (t ₉₀)	~6 seconds	To 1,000 ppm O ₂ under green LED
Recovery time	~9 minutes	After 2 min exposure to 1,000 ppm O ₂
Storage stability	>3 months	Dark storage, ambient conditions
Operational cycling stability	>6 hours continuous	Repeated exposure/recovery cycles
Humidity tolerance	0–80% RH	No measurable performance degradation
Selectivity	No response to N ₂ O, CO ₂ , H ₂ , CH ₄ , C ₂ H ₄ , CO, N ₂	Unique to O ₂ via photochemical mechanism
Target device resistance	1–5 kΩ	QC criterion for spray-coated Batch 2 electrodes
Sensitizer concentration (soaking)	0.2 mM in DMSO	Optimal from screening; 16h dark soak
LED parameters	~200 mW/mm ² green, 0.08A, 2.0 cm distance	LCFOCUS LC-10RGB-C30
TiO ₂ phase	Amorphous sol-gel	Outperforms anatase, P25, rutile at same loading
Excited-state potential E(S [*] /S ⁺)	–0.93 V vs NHE	Thermodynamic driver for O ₂ reduction

End of Document

*SWCNT–TiO₂–Re Sensor: From Supplementary Data to Distributed Manufacturing
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